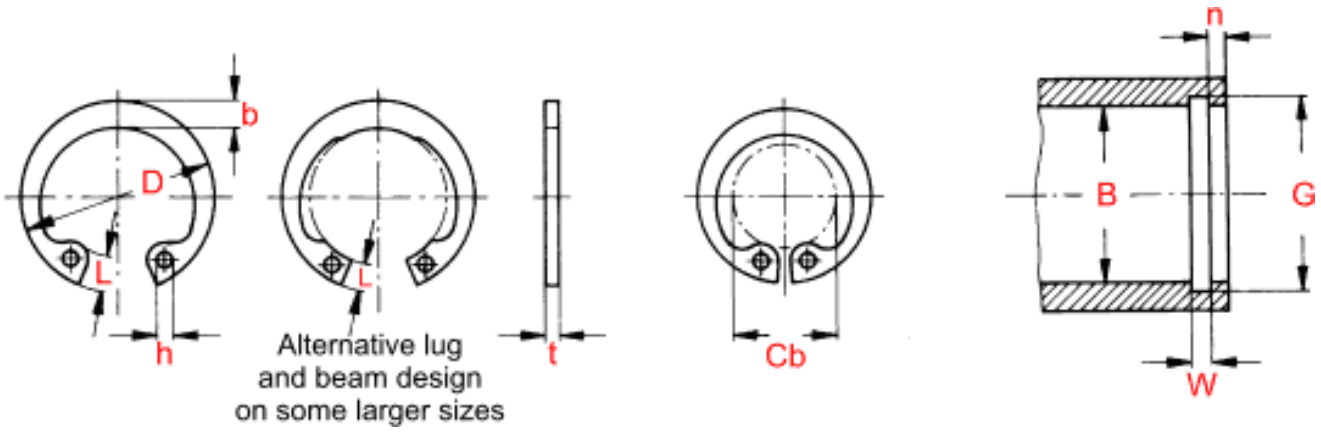
J C GUPTA & SONS

Standard Internal Circlips

Datasheet G2D22



Standard Internal Circlips: To DIN 472 Metric

The most common form of axially fitted circlips for bores with grooves. The circlip has a large free diameter and gap width, consequently it fits more tightly into the groove and provides higher and more uniform thrust load capacity. Lugs with holes are provided for rapid fitting or removal with circlip pliers.

Standard material: Carbon Spring Steel - Standard Finish: Phosphate and Oil.
(Standard circlips are also available in Stainless Steel)

Most sizes over 170mm are without lugs										Measurements in mm & kN					
Bore B	Circlip Dimensions									Groove Dimensions					PART NO.
	t	Tol	D	Tol	Cb	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
8	0.80	+0.00 -0.50	8.7	+0.36 -0.10	3.0	1.1	2.4	1.0	2.0	8.4	+0.09 -0.00	0.90	0.6	0.86	G2D2208000
9	0.80		9.8		3.7	1.3	2.5	1.0	2.0	9.4		0.90	0.6	0.96	G2D2209000
10	1.00		10.8		3.3	1.4	3.2	1.2	4.0	10.4	+0.11 -0.00	1.10	0.6	1.08	G2D2210000
11	1.00		11.8		4.1	1.5	3.3	1.2	4.0	11.4		1.10	0.6	1.17	G2D2211000
12	1.00		13.0		4.9	1.7	3.4	1.5	4.0	12.5		1.10	0.8	1.6	G2D2212000
13	1.00		14.1		5.4	1.8	3.6	1.5	4.2	13.6		1.10	0.9	2.1	G2D2213000
14	1.00		15.1		6.2	1.9	3.7	1.7	4.5	14.6		1.10	0.9	2.25	G2D2214000
15	1.00		16.2		7.2	2.0	3.7	1.7	5.0	15.7		1.10	1.1	2.8	G2D2215000
16	1.00		17.3		8.0	2.0	3.8	1.7	5.5	16.8		1.10	1.2	3.4	G2D2216000
17	1.00		18.3		8.8	2.1	3.9	1.7	6.0	17.8		1.10	1.2	3.6	G2D2217000
18	1.00		19.5		9.4	2.2	4.1	2.0	6.5	19.0		1.10	1.5	4.8	G2D2218000

19	1.00		20.5	+0.42 -0.13	10.4	2.2	4.1	2.0	6.8	20.0		1.10	1.5	5.1	G2D2219000
20	1.00		21.5		11.2	2.3	4.2	2.0	7.2	21.0	+0.13 -0.00	1.10	1.5	5.4	G2D2220000
21	1.00		22.5		12.2	2.4	4.2	2.0	7.6	22.0		1.10	1.5	5.7	G2D2221000
22	1.00		23.5		13.2	2.5	4.2	2.0	8.0	23.0		1.10	1.5	5.9	G2D2222000
23	1.20		24.6	+0.42 -0.21	14.2	2.5	4.2	2.0	13.8	24.1		1.30	1.5	6.8	G2D2223000
24	1.20		25.9		14.8	2.6	4.4	2.0	13.9	25.2		1.30	1.8	7.7	G2D2224000
25	1.20		26.9		15.5	2.7	4.5	2.0	14.6	26.2		1.30	1.8	8.0	G2D2225000
26	1.20		27.9		16.1	2.8	4.7	2.0	13.85	27.2	+0.21 -0.00	1.30	1.8	8.4	G2D2226000
27	1.20		29.1		17.1	2.9	4.7	2.0	13.3	28.4		1.30	2.1	10.1	G2D2227000
28	1.20	+0.00 -0.06	30.1	+0.50 -0.25	17.9	2.9	4.8	2.0	13.3	29.4		1.30	2.1	10.5	G2D2228000
29	1.20		31.1		18.4	3.0	4.8	2.0	13.6	30.4		1.30	2.1	10.9	G2D2229000
30	1.20		32.1		19.9	3.0	4.8	2.0	13.7	31.4		1.30	2.1	11.3	G2D2230000
31	1.20		33.4		20.0	3.2	5.2	2.5	13.8	32.7		1.30	2.6	14.1	G2D2231000
32	1.20		34.4		20.6	3.2	5.4	2.5	13.8	33.7		1.30	2.6	14.6	G2D2232000
33	1.20		35.5		21.6	3.3	5.4	2.5	14.3	34.7		1.30	2.6	15.0	G2D2233000
34	1.50		36.5		22.6	3.3	5.4	2.5	26.2	35.7		1.60	2.6	15.4	G2D2234000
35	1.50		37.8		23.6	3.4	5.4	2.5	26.9	37.0		1.60	3.0	18.8	G2D2235000
36	1.50		38.8		24.6	3.5	5.4	2.5	26.4	38.0		1.60	3.0	19.4	G2D2236000
37	1.50		39.8		25.4	3.6	5.5	2.5	27.1	39.0		1.60	3.0	19.8	G2D2237000
38	1.50		40.8		26.4	3.7	5.5	2.5	28.2	40.0	+0.25 -0.00	1.60	3.0	22.5	G2D2238000
39	1.50		42.0	+0.90 -0.39	27.2	3.8	5.6	2.5	28.8	41.0		1.60	3.0	26.0	G2D2239000
40	1.75		43.5		27.8	3.9	5.8	2.5	44.6	42.5		1.85	3.8	27.0	G2D2240000
41	1.75		44.5		28.6	4.0	5.9	2.5	45.0	43.5		1.85	3.8	27.6	G2D2241000
42	1.75		45.5		29.6	4.1	5.9	2.5	44.7	44.5		1.85	3.8	28.4	G2D2242000
43	1.75		46.5		30.6	4.2	5.9	2.5	44.5	45.5		1.85	3.8	28.8	G2D2243000
44	1.75		47.5		31.4	4.2	6.0	2.5	43.3	46.5		1.85	3.8	29.5	G2D2244000
45	1.75		48.5		32.0	4.3	6.2	2.5	43.1	47.5		1.85	3.8	30.2	G2D2245000
46	1.75		49.5		32.7	4.4	6.3	2.5	42.9	48.5		1.85	3.8	30.8	G2D2246000
47	1.75		50.5	+1.10 -0.46	33.5	4.4	6.4	2.5	43.5	49.5		1.85	3.8	31.4	G2D2247000
48	1.75		51.5		34.5	4.5	6.4	2.5	43.2	50.5		1.85	3.8	32.0	G2D2248000
50	2.00		54.2		36.3	4.6	6.5	2.5	60.8	53.0		2.15	4.5	40.5	G2D2250000
51	2.00		55.2		37.3	4.7	6.5	2.5	60.2	54.0	+0.30 -0.00	2.15	4.5	41.2	G2D2251000
52	2.00	+0.00 -0.07	56.2		37.9	4.7	6.7	2.5	60.25	55.0		2.15	4.5	42.0	G2D2252000
53	2.00		57.2		38.9	4.9	6.7	2.5	60.7	56.0		2.15	4.5	42.9	G2D2253000
54	2.00		58.2		39.9	5.0	6.7	2.5	60.4	57.0		2.15	4.5	43.6	G2D2254000

Most sizes over 170mm are without lugs

Measurements in mm & kN

Key to table headings and abbreviations

Bore B	Circlip Dimensions									Groove Dimensions					PART NO.
	t	Tol	D	Tol	Cb	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
55	2.00	+0.00 -0.07	59.2	+1.10 -0.46	40.7	5.0	6.8	2.5	60.3	58.0	+0.30 -0.00	2.15	4.5	44.4	G2D2255000
56	2.00		60.2		41.7	5.1	6.8	2.5	60.3	59.0		2.15	4.5	45.2	G2D2256000
57	2.00		61.2		42.7	5.1	6.8	2.5	60.8	60.0		2.15	4.5	46.0	G2D2257000
58	2.00		62.2		43.5	5.2	6.9	2.5	60.8	61.0		2.15	4.5	46.7	G2D2258000
60	2.00		64.2		44.7	5.4	7.3	2.5	61.0	63.0		2.15	4.5	48.3	G2D2260000
62	2.00		66.2		46.7	5.5	7.3	2.5	60.9	65.0		2.15	4.5	49.8	G2D2262000
63	2.00		67.2		47.7	5.6	7.3	2.5	60.8	66.0		2.15	4.5	50.6	G2D2263000
64	2.00		68.2		48.2	5.7	7.5	2.5	60.6	67.0		2.15	4.5	51.4	G2D2264000
65	2.50		69.2		49.0	5.8	7.6	3.0	121.0	68.0		2.65	4.5	51.8	G2D2265000
67	2.50		71.5		50.8	6.0	7.7	3.0	121.0	70.0		2.65	4.5	53.8	G2D2267000
68	2.50		72.5		51.6	6.1	7.8	3.0	121.5	71.0		2.65	4.5	54.5	G2D2268000
70	2.50		74.5		53.6	6.2	7.8	3.0	119.0	73.0		2.65	4.5	56.2	G2D2270000
72	2.50		76.5		55.6	6.4	7.8	3.0	119.2	75.0		2.65	4.5	58.0	G2D2272000
75	2.50		79.5		58.6	6.6	7.8	3.0	118.0	78.0		2.65	4.5	60.0	G2D2275000
77	2.50		81.5		60.4	6.7	7.9	3.0	121.1	80.0		2.65	4.5	61.6	G2D2277000
78	2.50		82.5	+1.30 -0.54	60.1	6.8	8.5	3.0	122.5	81.0	+0.35 -0.00	2.65	4.5	62.3	G2D2278000
80	2.50		85.5		62.1	7.0	8.5	3.0	120.9	83.5		2.65	5.3	74.6	G2D2280000
82	2.50		87.5		64.1	7.0	8.5	3.0	119.0	85.5		2.65	5.3	76.6	G2D2282000
85	3.00	+0.00 -0.08	90.5		66.9	7.2	8.6	3.5	201.4	88.5		3.15	5.3	79.5	G2D2285000
87	3.00		92.5		68.9	7.3	8.6	3.5	204.5	90.5		3.15	5.3	81.3	G2D2287000
88	3.00		93.5		69.9	7.4	8.6	3.5	209.4	91.5		3.15	5.3	82.1	G2D2288000
90	3.00		95.5		71.9	7.6	8.6	3.5	199.0	93.5		3.15	5.3	84.0	G2D2290000
92	3.00		97.5		73.7	7.8	8.7	3.5	201.0	95.5		3.15	5.3	85.8	G2D2292000
95	3.00		100.5		76.5	8.1	8.8	3.5	195.0	98.5		3.15	5.3	88.6	G2D2295000
97	3.00		102.5		78.5	8.2	8.8	3.5	193.0	100.5		3.15	5.3	90.5	G2D2297000
98	3.00		103.5		79.0	8.3	9.0	3.5	191.0	101.5		3.15	5.3	91.3	G2D2298000
100	3.00		105.5		80.6	8.4	9.2	3.5	188.0	103.5		3.15	5.3	93.1	G2D2210000
102	4.00		108.0		82.0	8.5	9.5	3.5	439.0	106.0	+0.54	4.15	6.0	108.8	G2D2210200
105	4.00		112.0		85.0	8.7	9.5	3.5	436.0	109.0		4.15	6.0	112.0	G2D2210500
108	4.00		115.0		88.0	8.9	9.5	3.5	419.0	112.0		4.15	6.0	115.0	G2D2210800

110	4.00		117.0		88.2	9.0	10.4	3.5	415.0	114.0	-0.00	4.15	6.0	117.0	G2D2211000
112	4.00		119.0		90.0	9.1	10.5	3.5	418.0	116.0		4.15	6.0	119.0	G2D2211200
115	4.00		122.0		93.0	9.3	10.5	3.5	409.0	119.0		4.15	6.0	122.0	G2D2211500
120	4.00		127.0		96.9	9.7	11.0	3.5	396.0	124.0		4.15	6.0	127.0	G2D2212000
125	4.00		132.0		101.9	10.0	11.0	4.0	385.0	129.0		4.15	6.0	132.0	G2D2212500
130	4.00		137.0		106.9	10.2	11.0	4.0	374.0	134.0		4.15	6.0	138.0	G2D2213000
135	4.00		142.0		111.5	10.5	11.2	4.0	358.0	139.0		4.15	6.0	143.0	G2D2213500
140	4.00	+0.00	147.0	+1.50	116.5	10.7	11.2	4.0	350.0	144.0		4.15	6.0	148.0	G2D2214000
145	4.00	-0.10	152.0	-0.63	121.0	10.9	11.4	4.0	336.0	149.0	+0.63	4.15	6.0	153.0	G2D2214500
150	4.00		158.0		124.8	11.2	12.0	4.0	326.0	155.0	-0.00	4.15	7.5	191.0	G2D2215000
155	4.00		164.0		129.8	11.4	12.0	4.0	324.0	160.0		4.15	7.5	206.0	G2D2215500
160	4.00		169.0		132.7	11.6	13.0	4.0	321.0	165.0		4.15	7.5	212.0	G2D2216000
165	4.00		174.5		137.7	11.8	13.0	4.0	319.0	170.0		4.15	7.5	219.0	G2D2216500
170	4.00		179.5		141.6	12.2	13.5	4.0	349.0	175.0		4.15	7.5	225.0	G2D2217000
175	4.00		184.5		146.6	12.7	13.5	4.0	351.0	180.0		4.15	7.5	232.0	G2D2217500
180	4.00		189.5		150.2	13.2	14.2	4.0	347.0	185.0		4.15	7.5	238.0	G2D2218000
185	4.00		194.5	+1.70	155.2	13.7	14.2	4.0	349.0	190.0		4.15	7.5	245.0	G2D2218500
190	4.00		199.5	-0.72	160.2	13.8	14.2	4.0	340.0	195.0	+0.72	4.15	7.5	251.0	G2D2219000
195	4.00		204.5		165.2	13.8	14.2	4.0	330.0	200.0	-0.00	4.15	7.5	258.0	G2D2219500
200	4.00		209.5		170.2	14.0	14.2	4.0	325.0	205.0		4.15	7.5	265.0	G2D2220000

Most sizes over 170mm are without lugs

Measurements in mm & kN

Bore B	Circlip Dimensions									Groove Dimensions					PART NO.
	t	Tol	D	Tol	Cb	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
205	5.00		217.0		175.2	14.0	14.2	4.0	616.0	211.0		5.15	9.0	326.0	G2D2220500
210	5.00		222.0		180.2	14.0	14.2	4.0	601.0	216.0		5.15	9.0	333.0	G2D2221000
215	5.00		227.0		185.2	14.0	14.2	4.0	586.0	221.0		5.15	9.0	341.0	G2D2221500
220	5.00		232.0	+1.70	190.2	14.0	14.2	4.0	574.0	226.0		5.15	9.0	349.0	G2D2222000
225	5.00		237.0	-0.72	195.2	14.0	14.2	4.0	560.0	231.0	+0.72	5.15	9.0	357.0	G2D2222500
230	5.00		242.0		200.2	14.0	14.2	4.0	549.0	236.0	-0.00	5.15	9.0	365.0	G2D2223000
235	5.00		247.0		205.2	14.0	14.2	4.0	536.0	241.0		5.15	9.0	373.0	G2D2223500
240	5.00		252.0		210.2	14.0	14.2	4.0	525.0	246.0		5.15	9.0	380.0	G2D2224000
245	5.00		257.0		215.2	14.0	14.2	4.0	514.0	251.0		5.15	9.0	389.0	G2D2224500
250	5.00	+0.00	262.0		220.2	14.0	14.2	4.0	504.0	256.0		5.15	9.0	396.0	G2D2225000
255	5.00	-0.12	270.0		221.0	16.0	16.2	5.0	549.0	263.0		5.15	12.0	541.0	G2D2225500

260	5.00		275.0	+2.00 -0.81	226.0	16.0	16.2	5.0	538.0	268.0	+0.81 -0.00	5.15	12.0	553.0	G2D2226000
265	5.00		280.0		231.0	16.0	16.2	5.0	528.0	273.0		5.15	12.0	563.0	G2D2226500
270	5.00		285.0		236.0	16.0	16.2	5.0	518.0	278.0		5.15	12.0	573.0	G2D2227000
275	5.00		290.0		241.0	16.0	16.2	5.0	509.0	283.0		5.15	12.0	585.0	G2D2227500
280	5.00		295.0		246.0	16.0	16.2	5.0	499.0	288.0		5.15	12.0	593.0	G2D2228000
285	5.00		300.0		251.0	16.0	16.2	5.0	491.0	293.0		5.15	12.0	605.0	G2D2228500
290	5.00		305.0		256.0	16.0	16.2	5.0	482.0	298.0		5.15	12.0	615.0	G2D2229000
295	5.00		310.0		261.0	16.0	16.2	5.0	474.0	303.0		5.15	12.0	625.0	G2D2229500
300	5.00		315.0		266.0	16.0	16.2	5.0	466.0	308.0		5.15	12.0	636.0	G2D2230000
310	6.00	+0.00 -0.18	327.0	+2.50 -1.00	268.0	20.0	20.2	6.0	947.0	320.0	+0.89 -0.00	6.20	15.0	823.0	G2D2231000
320	6.00		337.0		278.0	20.0	20.2	6.0	919.0	330.0		6.20	15.0	850.0	G2D2232000
330	6.00		347.0		288.0	20.0	20.2	6.0	894.0	340.0		6.20	15.0	876.0	G2D2233000
340	6.00		357.0		298.0	20.0	20.2	6.0	869.0	350.0		6.20	15.0	903.0	G2D2234000
350	6.00		367.0		308.0	20.0	20.2	6.0	846.0	360.0		6.20	15.0	929.0	G2D2235000
360	6.00		377.0		318.0	20.0	20.2	6.0	823.0	370.0		6.20	15.0	955.0	G2D2236000
370	6.00		387.0		328.0	20.0	20.2	6.0	803.0	380.0		6.20	15.0	981.0	G2D2237000
380	6.00		397.0		338.0	20.0	20.2	6.0	784.0	390.0		6.20	15.0	1008.0	G2D2238000
390	6.00		407.0		348.0	20.0	20.2	6.0	764.0	400.0		6.20	15.0	1033.0	G2D2239000
400	6.00		417.0		358.0	20.0	20.2	6.0	746.0	410.0	+1.00 -0.00	6.20	15.0	1060.0	G2D2240000